Quadratics & Nonlinear Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $\sqrt{2x+1}=5$, what is x?

Answer: 12. Applying Equivalent Expressions & Quadratic Equations principles, $2x+1=25 \Rightarrow x=12$. Therefore, the verified correct option is 12.

2 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In the plane, a parabola is given by $f(x)=3(x-4)^2+11$. What is the vertical intercept $f(0)$?

Answer: -1. Setting $x=0$ gives $f(0)=3(-4)^2+11=59$.

3 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=2x^2+x-7$, what is $f(6)$?

Answer: 71. Applying Equivalent Expressions & Quadratic Equations principles, $f(6)=2(6)^2+1(6)-7=71$ Therefore, the verified correct option is 71.

4 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=2x^2+2x-3$, what is $f(2)$?

Answer: 9. Applying Equivalent Expressions & Quadratic Equations principles, $f(2)=2(2)^2+2(2)-3=9$ Therefore, the verified correct option is 9.

5 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=x^2-6x-2$, what is $f(5)$?

Answer: -7. Applying Equivalent Expressions & Quadratic Equations principles, $f(5)=1(5)^2-6(5)-2=-7$ Therefore, the verified correct option is -7.

6 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

An exponential growth model is defined by $f(x)=171\cdot 2^{x/9}$. What is the value $f(18)$?

Answer: 87. Evaluating at $x=18$ gives $171\cdot 2^2=684$.

7 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The product of two consecutive positive integers is 56. What is the smaller integer?

Answer: 7. $n(n+1)=56 \Rightarrow n^2+n-56=0 \Rightarrow (n-7)(n+8)=0 \Rightarrow n=7$

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Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph shows $f(x) = -x^2 + 4$. What is the maximum value of f?

Answer: 4. The vertex of $f(x) = -x^2 + 4$ is at $(0, 4)$, which is the maximum value.

9 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2 - 4x - 32 = 0$?

Answer: 8. Factor: $(x + 4)(x - 8) = 0 \Rightarrow x = -4$ or $x = 8$. Larger $= 8$.

10 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 2x^2 + 2x - 2$, what is $f(5)$?

Answer: 58. Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 2(5)^2 + 2(5) - 2 = 58$ Therefore, the verified correct option is 58.

11 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 4x^2 + 2x + 7$, what is $f(-4)$?

Answer: 63. Applying Equivalent Expressions & Quadratic Equations principles, $f(-4) = 4(-4)^2 + 2(-4) + 7 = 63$ Therefore, the verified correct option is 63.

12 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The parabola $y = x^2 - 8x + 15$ intersects the x-axis at $x = 3$ and $x = 5$. What is the x-coordinate of the vertex?

Answer: 4. Vertex is midway: $x = \frac{3+5}{2} = 4$.

13 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For what sum of x-values does the graph of $y = x^2 - 3x - 10$ cross the x-axis?

Answer: 3. The x-intercepts are $x = -2$ and $x = 5$ (from factoring), so their sum is 3.

14 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the sum of the solutions to $x^2 + 5x - 36 = 0$?

Answer: -5. By Vieta's formulas, sum of roots $= -\frac{b}{a} = -\frac{5}{1} = -5$.

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Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2 + 5x + 6 = 0$?

Answer: -2. Factor: $(x + 2)(x + 3) = 0 \Rightarrow x = -2$ or $x = -3$. Larger $= -2$.

16 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. What is the remainder $f(2)$ upon division by $(x - 2)$?

Answer: 3. By the Remainder Theorem, $f(2) = 2^3 - 5(2^2) + 10(2) - 20 = -12$.

17 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The quadratic function $f(x) = -x^2 + 12x - 18$ attains its maximum at $x = g$. What is the maximum value $f(g)$?

Answer: 107. The vertex occurs at $g = 6$, giving $f(6) = 36 - 18 = 18$.

18 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - 5x + 5$, what is $f(-4)$?

Answer: 41. Applying Equivalent Expressions & Quadratic Equations principles, $f(-4) = 1(-4)^2 - 5(-4) + 5 = 41$ Therefore, the verified correct option is 41.

19 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x^2 + 6x + 9 = 25$, what is a possible value of $x + 3$?

Answer: -5. $(x + 3)^2 = 25 \Rightarrow x + 3 = 5$ or $x + 3 = -5$

20 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In the partial fraction identity $\frac{6}{(x-11)(x-13)} = \frac{f}{x-13} - \frac{g}{x-11}$, f and g are constants. What is the sum $f + g$?

Answer: 2. Matching numerators gives $f = 3$ and $g = 3$, so $f + g = 6$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Which value of x satisfies $x^2 - 5x + 6 = 0$?

a. $x = 1$

b. $x = 2$ ✓

c. $x = 4$

d. $x = 6$

$(x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$. Choice: $x = 2$.

22 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -2x^2 - 32x - 136$?

a. (-7, -8)

b. (8, -8)

c. (-8, 8)

d. **(-8, -8)** ✓

Vertex $x = -\frac{b}{2a} = -8$, $y = f(-8) = -8$. Vertex $= (-8, -8)$.

23 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = 2x^2 + 2x$ have?

a. Zero

b. One

c. **Two** ✓

d. Cannot be determined

Discriminant $= b^2 - 4ac = (2)^2 - 4(2)(0) = 4$, which is positive, giving two real intercept(s).

24 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = 2x^2 + 6x$ have?

a. Zero

b. One

c. **Two** ✓

d. Cannot be determined

Discriminant $= b^2 - 4ac = (6)^2 - 4(2)(0) = 36$, which is positive, giving two real intercept(s).

25 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = x^2 - 8x + 21$?

a. (5, 5)

b. **(4, 5)** ✓

c. (4, -5)

d. (-4, 5)

Vertex $x = -\frac{b}{2a} = 4$, $y = f(4) = 5$. Vertex $= (4, 5)$.

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26 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = x^2 - 2x + 4$ have?

a. Zero ✓

b. One

c. Two

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-2)^2 - 4(1)(4) = -12$, which is negative, giving zero real intercept(s).

27 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 + 4x - 4$ have?

a. Zero

b. One ✓

c. Two

d. Cannot be determined

Discriminant $= b^2 - 4ac = (4)^2 - 4(-1)(-4) = 0$, which is zero, giving one real intercept(s).

28 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 + 5x - 1$ have?

a. Zero

b. One

c. Two ✓

d. Cannot be determined

Discriminant $= b^2 - 4ac = (5)^2 - 4(-1)(-1) = 21$, which is positive, giving two real intercept(s).

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What are the solutions to $x^2 - 5x + 6 = 0$?

a. $x = 1, 6$

b. $x = 2, 3$ ✓

c. $x = -2, -3$

d. $x = -1, 6$

$(x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$.

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30 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -3x^2 - 42x - 150$?

a. (-7, 3)

b. **(-7, -3)** ✓

c. (-6, -3)

d. (7, -3)

Vertex $x = -\frac{b}{2a} = -7$, $y = f(-7) = -3$. Vertex $= (-7, -3)$.

31 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -2x^2 - 5x - 3$ have?

a. Zero

b. One

c. **Two** ✓

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-5)^2 - 4(-2)(-3) = 1$, which is positive, giving two real intercept(s).

32 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 4x^2 + 7x$, what is $f(5)$?

a. 135 ✓

b. 138

c. 131

d. 139

Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 4(5)^2 + 7(5) = 135$ Therefore, the verified correct option is 135.

33 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 5x^2 + 3$, what is $f(4)$?

a. 77

b. 78

c. 76

d. 83 ✓

Applying Equivalent Expressions & Quadratic Equations principles, $f(4) = 5(4)^2(4) + 3 = 83$ Therefore, the verified correct option is 83.

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34 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -3x^2 - 6x + 5$?

a. **(-1, 8)** ✓

b. (-1, -8)

c. (1, 8)

d. (0, 8)

Vertex $x = -\frac{b}{2a} = -1$, $y = f(-1) = 8$. Vertex $= (-1, 8)$.

35 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 4x^2 - x + 9$, what is $f(1)$?

a. 17

b. 20

c. 12 ✓

d. 8

Applying Equivalent Expressions & Quadratic Equations principles, $f(1) = 4(1)^2 - 1(1) + 9 = 12$

Therefore, the verified correct option is 12.

36 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 3x^2 - x + 7$, what is $f(-5)$?

a. 87 ✓

b. 82

c. 80

d. 94

Applying Equivalent Expressions & Quadratic Equations principles, $f(-5) = 3(-5)^2 - 1(-5) + 7 = 87$

Therefore, the verified correct option is 87.

37 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the y-intercept of the parabola given by $y = 3x^2 - 7x + 12$?

a. $(0, 3)$

b. $(0, -7)$

c. $(0, 12)$ ✓

d. $(12, 0)$

Evaluate at $x = 0$: $y = 3(0)^2 - 7(0) + 12 = 12$. The y-intercept is $(0, 12)$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Which is equivalent to $(x + 4)^2 - 9$?

a. $x^2 + 8x + 7$ ✓

b. $x^2 + 8x - 9$

c. $x^2 + 16x + 7$

d. $x^2 + 4x + 7$

Applying Equivalent Expressions & Quadratic Equations principles, $x^2 + 8x + 16 - 9 = x^2 + 8x + 7$. Therefore, the verified correct option is $x^2 + 8x + 7$.

39 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = x^2 - 14x + 53$?

a. (7, -4)

b. (-7, 4)

c. **(7, 4)** ✓

d. (8, 4)

Vertex $x = -\frac{b}{2a} = 7$, $y = f(7) = 4$. Vertex $= (7, 4)$.

40 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 - x + 2$ have?

a. Zero

b. One

c. **Two** ✓

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-1)^2 - 4(-1)(2) = 9$, which is positive, giving two real intercept(s).

41 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 2x^2 + 3x + 5$, what is $f(1)$?

a. 10 ✓

b. 11

c. 15

d. 4

Applying Equivalent Expressions & Quadratic Equations principles, $f(1) = 2(1)^2 + 3(1) + 5 = 10$ Therefore, the verified correct option is 10.

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42 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2 + x - 72 = 0$?

a. 15

b. 14

c. 8 ✓

d. 1

Factor: $(x + 9)(x - 8) = 0 \Rightarrow x = -9$ or $x = 8$. Larger $= 8$.

43 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = 2x^2 + 24x + 78$?

a. (-5, 6)

b. **(-6, 6)** ✓

c. (6, 6)

d. (-6, -6)

Vertex $x = -\frac{b}{2a} = -6$, $y = f(-6) = 6$. Vertex $= (-6, 6)$.

44 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - 8x + 6$, what is $f(5)$?

a. -9 ✓

b. -15

c. -14

d. -18

Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 1(5)^2 - 8(5) + 6 = -9$

Therefore, the verified correct option is -9.

45 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - x - 8$, what is $f(6)$?

a. 27

b. 31

c. 22 ✓

d. 28

Applying Equivalent Expressions & Quadratic Equations principles, $f(6) = 1(6)^2 - 1(6) - 8 = 22$

Therefore, the verified correct option is 22.

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